

Construction of sustainable tourism model from the perspective of dynamic multi-objective optimization

Summary

The city of Juneau, Alaska, recorded 1.6 million cruise passengers in 2023, contributing \$375 million in revenue but causing significant environmental and infrastructure challenges. Overtourism has impacted the Mendenhall Glacier, which has receded by the equivalent of eight football fields since 2007. This highlights the urgent need for a sustainable tourism model that balances revenue generation with environmental preservation and resident satisfaction.

For Task 1, we developed a **dynamic multi-objective model** for sustainable tourism. Key factors include tourist volume, revenue, environmental quality, infrastructure capacity, and resident satisfaction. The model utilizes a system of **differential equations** to capture interactions among these factors, with **optimization objectives** to maximize tourism revenue while minimizing environmental and infrastructure pressures. **Constraints** ensure environmental quality, infrastructure health, and resident satisfaction remain above critical thresholds. **Feedback mechanisms** allocate a portion of tax revenue to sustainability measures, enhancing long-term capacity.

For Task 2, we adapted the model for other destinations by **categorizing** them as environmentally sensitive, cultural heritage sites, or urban leisure destinations. Parameters were adjusted to reflect local characteristics, such as environmental sensitivity coefficients and infrastructure thresholds. A **scoring system** prioritizes lesser-known attractions to distribute tourist flows more evenly. Case studies, such as Iceland, demonstrate how measures like dynamic tax policies and attraction recommendations can mitigate overtourism.

Sensitivity analysis identified key parameters, including environmental sensitivity and tax rates, as critical to model outcomes. The results show that active management stabilizes environmental and infrastructure conditions while sustaining revenue growth. Our model provides a flexible framework for promoting sustainable tourism across diverse destinations.

Finally, a **memorandum** was prepared for the Juneau Tourism Council, summarizing our findings, providing actionable recommendations, and emphasizing the importance of balancing tourism growth with sustainability efforts.

Keywords: Sustainable tourism, differential equations, multi-objective optimization, environmental preservation, infrastructure capacity.

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1 Introduction

1.1 Background

The city of Juneau in Alaska, USA, is a highly popular tourist destination, particularly renowned for its cruise tourism industry. In 2023, the city welcomed 1.6 million cruise tourists, with visitor numbers reaching as high as 20,000 on the busiest days. While tourism has brought in a substantial revenue of \$375 million, it has also led to environmental degradation, infrastructure strain, and dissatisfaction among local residents. One of the most severe impacts is on the Mendenhall Glacier, one of Juneau's main attractions, which has significantly receded due to climate change and overtourism. Although the local government has implemented several measures, including increasing hotel taxes, visitor fees, daily visitor caps, and restrictions on alcohol sales and consumption, the development and implementation of a sustainable tourism plan remains an urgent issue.[1] Such a plan would ensure that tourists can enjoy their vacations without incurring excessive costs, while also preserving the local environmental quality and the well-being of indigenous residents.



Figure 1: Juneau city scenery[2]

1.2 Restatement and Analysis of the Problem

We need to complete the following tasks based on the data provided in the prompt and additional data we have collected:

Task 1: Construct a model to provide sustainable recommendations for local tourism development. This model should consider the number of tourists, total revenue, and measures implemented to stabilize the tourism industry. The model should explicitly define the factors to be optimized and the constraints to be respected. It should also include a plan for how to allocate additional revenue expenditures and demonstrate how these expenditures can be fed back into the model to promote sustainable tourism.

Task 2: Explore how to apply the aforementioned model to other destinations impacted by overtourism, and design a mechanism to promote lesser-known attractions in other regions.

Task 3: Prepare a memorandum to the Juneau Tourism Committee summarizing our forecasts, the potential impacts of various measures, and recommendations for optimizing results.

1.3 Overview of our work

Based on the above analysis, we have undertaken the following tasks. Our working framework is illustrated in the figure below.

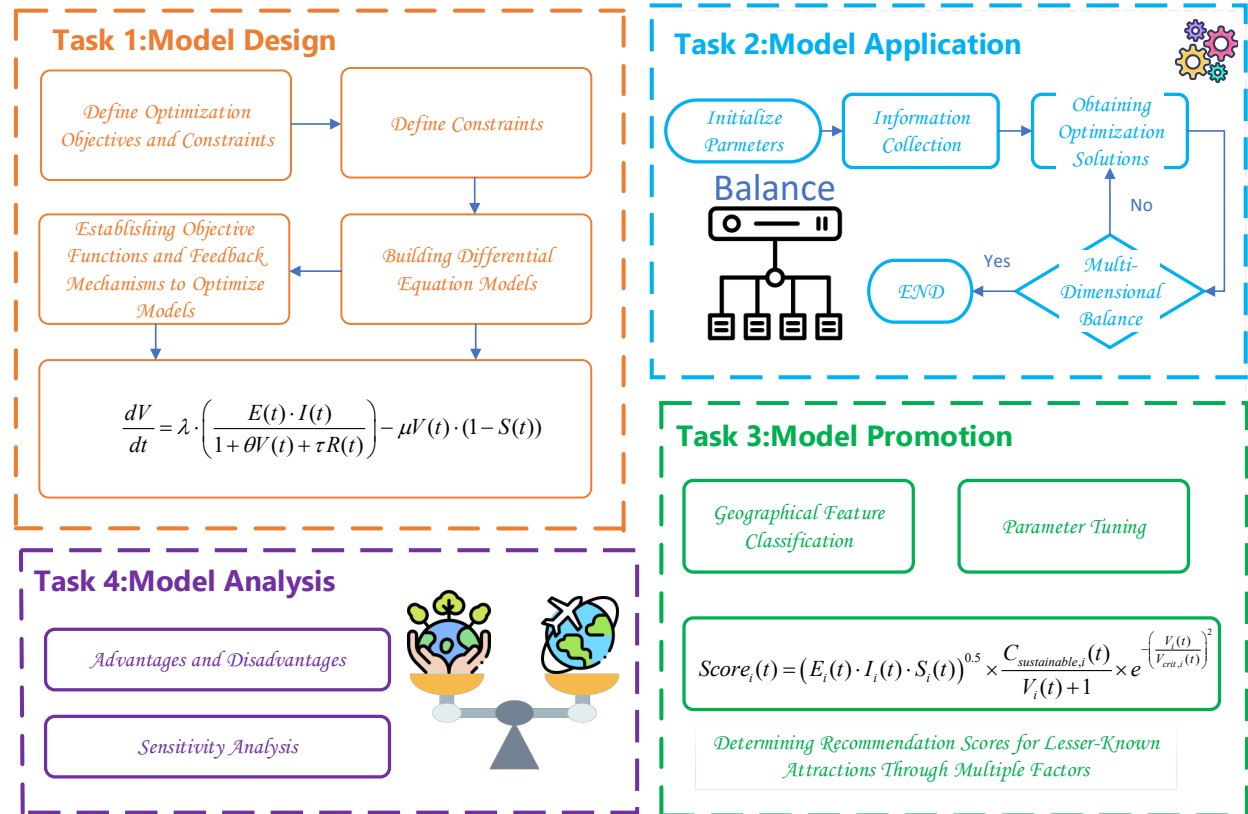


Figure 2: The Framework of Our Work

The detailed modeling process and model specifics will be presented in the following sections.

2 Assumptions and Justifications

To simplify the problem and facilitate the design of a reasonable sustainable tourism model, we have proposed the following basic assumptions, each of which is rational and justifiable.

Assumption 1: Relationship between Tourist Numbers and Revenue: It is assumed that there is a positive correlation between the number of tourists and tourism revenue, but with diminishing marginal returns. In other words, as the number of tourists increases, the additional revenue generated by each additional tourist gradually decreases.

Assumption 2: Infrastructure Capacity: It is assumed that local infrastructure (such as transportation, accommodation, waste management, etc.) has a maximum capacity. Exceeding this capacity would lead to a decline in service quality and a reduction in residents' quality of life.

Assumption 3: Environmental Carrying Capacity: It is assumed that the environment has a maximum carrying capacity, beyond which irreversible damage may occur. This carrying capacity can be estimated using historical data and environmental studies.

Assumption 4: Resident Satisfaction: It is assumed that resident satisfaction is inversely proportional to the number of tourists. As the number of tourists increases, resident satisfaction decreases, and vice versa.

Assumption 5: Effectiveness of Measures: It is assumed that the measures implemented will directly impact tourist numbers, revenue, and resident satisfaction. For example, increasing taxes may reduce the number of tourists but increase revenue per tourist.

Assumption 6: Feedback Effect of Expenditures: It is assumed that expenditures on environmental protection and infrastructure improvements will enhance environmental carrying capacity and infrastructure capacity, thereby allowing for more tourists while simultaneously increasing resident satisfaction.

3 Notations

Symbols	Description
$V(t)$	The number of tourists at time t
$E(t)$	Environmental quality at time t
$I(t)$	Infrastructure health at time t
$S(t)$	Resident satisfaction at time t
$R(t)$	Total tourism revenue at time t
$C(t)$	Sustainable expenditure at time t
$N(t)$	Normal revenue at time t
$T(t)$	Tax revenue at time t
$h(t)$	Tax rate at time t
k_E	Environmental sensitivity coefficient
α_I	Infrastructure sensitivity coefficient
S_{min}	Minimum resident satisfaction threshold
λ	Tourist growth rate coefficient
α_E	Environmental restoration coefficient
γ_S	Resident satisfaction sensitivity coefficient

4 Task 1: Sustainable Tourism Dynamic Multi-Objective

Model

4.1 Factors to consider

When constructing a dynamic multi-objective model for sustainable tourism, we comprehensively considered the following core factors and their mathematical relationships to quantify the dynamic impact of tourist numbers on the economy, environment, infrastructure, and resident satisfaction:

Relationship between Normal Revenue (N) and Tourist Numbers (V):

Tourism revenue is a core economic indicator in the model. We assume that tourism revenue has a nonlinear relationship with tourist numbers, following the law of diminishing marginal returns. This means that as the number of tourists increases, the incremental revenue generated

by each additional tourist gradually decreases. The mathematical expression for this relationship is: $N = aV - bV^2$.

Relationship between Tax Revenue (T) and Normal Revenue (N):

Tax revenue represents the contribution of tourism to local finances, and it is assumed to be proportional to normal revenue: $T = gN$.

Feedback Mechanism between Sustainable Expenditure (C) and Total Revenue (R):

Sustainable expenditure is a fixed proportion of total revenue, allocated to environmental protection, infrastructure improvement, and community projects, forming a dynamic feedback loop: $C = hR$.

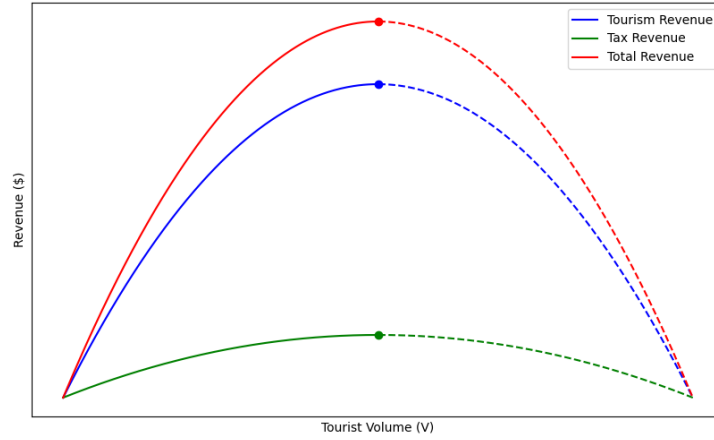


Figure 3: Trends in Tourism Revenue, Tax Revenue and Total Revenue as a function of Tourist Volume

4.2 Optimization objectives and constraints

In this section, we will introduce the optimization objectives and constraints of our model:

Optimization Objectives:

Maximize Total Tourism Revenue $R(t)$:

Tourism revenue is a core economic indicator of the tourism industry. We assume that tourism revenue has a nonlinear relationship with tourist numbers, with diminishing marginal returns. The objective is to maximize tourism revenue by optimizing the number of tourists $V(t)$:

$$\text{Maximize } R(t) = (1 + g)(aV(t) - bV(t)^2) \quad (1)$$

Minimize Environmental Impact $E(t)$:

Environmental impact measures the negative effects of tourism on the natural environment. It is a critical indicator of whether tourism in the region can be sustained in the long term. Our goal is to minimize environmental impact to ensure the sustainable development of local tourism:

$$\text{Minimize } E(t) = \frac{cV(t)}{E_{\max}} \quad (2)$$

Minimize Infrastructure Pressure $I(t)$:

Infrastructure pressure reflects the burden imposed by tourism on local infrastructure, including transportation, water supply, waste management, and more. Reducing infrastructure pressure is one of the core factors in ensuring the sustainable development of tourism in Juneau. Our goal is to minimize infrastructure pressure:

$$\text{Minimize } I(t) = \frac{dV(t)}{I_{\max}} \quad (3)$$

Constraints:

Environmental Quality Index Constraint:

The environmental quality index must not fall below the minimum threshold. Otherwise, it will lead to irreversible environmental damage and severe consequences. Therefore, the constraint for environmental impact is:

$$E(t) \geq E_{\min} \quad (4)$$

Infrastructure Health Index Constraint:

The infrastructure health index must not fall below the minimum threshold. Otherwise, it will result in a decline in service quality and a reduction in residents' quality of life. Therefore, the constraint for infrastructure pressure is:

$$I(t) \geq I_{\min} \quad (5)$$

Resident Satisfaction Constraint:

Resident satisfaction must not fall below the minimum required level. Otherwise, it will lead to resident dissatisfaction and social issues. Therefore, the constraint for resident satisfaction is:

$$S(t) \geq S_{min} \quad (6)$$

Non-Negative Tourist Number Constraint:

The number of tourists must be non-negative and less than a specified threshold:

$$0 \leq V(t) \leq V_{max} \quad (7)$$

4.3 Differential equation model

4.3.1 Establishment of Differential Equations

To capture the dynamic behavior of the system, we introduce the time variable t and establish a set of differential equations to describe the changes in tourist numbers, environmental quality, infrastructure health, and resident satisfaction. These equations account for the interactions between economic, environmental, and social factors, providing a framework to analyze and optimize sustainable tourism development.

1. Tourist Number Dynamics:

The change in tourist numbers $V(t)$ is influenced by promotional efforts, environmental quality, infrastructure, and resident attitudes. The dynamic equation for tourist numbers is:

$$\frac{dV}{dt} = \lambda \cdot \left(\frac{E(t) \cdot I(t)}{1 + \theta V(t) + \tau R(t)} \right) - \mu V(t) \cdot (1 - S(t)) \quad (8)$$

2. Environmental Quality Dynamics:

The change in environmental quality $E(t)$ is affected by both tourist-induced damage and restoration measures.[3] The dynamic equation for environmental quality is:

$$\frac{dE}{dt} = r_E \times (E(t) - E_{min}) - k_E \times V(t) + \alpha_E \times C_{env}(t) \quad (9)$$

3. Infrastructure Health Dynamics:

The change in infrastructure health $I(t)$ is influenced by tourist-induced wear and maintenance investments.[4] The dynamic equation for infrastructure health is:

$$\frac{dI}{dt} = -r_I \cdot (I(t) - I_{min}) - k_I \cdot V(t) + \alpha_I \cdot C_{infra}(t) \quad (10)$$

4. Resident Satisfaction Dynamics:

The change in resident satisfaction $S(t)$ is influenced by tourist numbers, income distribution, and community projects.[5] The dynamic equation for resident satisfaction is:

$$\frac{dS}{dt} = \gamma_S \cdot C_{comm}(t) - c_S \cdot V(t) \quad (11)$$

4.3.2 Restatement of the Optimization Function

To optimize multiple objectives simultaneously, we combine multiple objective functions into a single objective function using the weighting method. By applying different weights to different single optimization problems, we distinguish the importance of each problem within the overall optimization problem.[6] The final optimization problem can be expressed as:

$$\max \int_0^T [w_R R(t) - w_E (1 - E(t)) - w_I (1 - I(t))] dt \quad (w_R + w_E + w_I = 1) \quad (12)$$

$$s.t. \begin{cases} E(t) \geq E_{min} \\ I(t) \geq I_{min} \\ S(t) \geq S_{min} \\ 0 \leq V(t) \leq V_{max} \end{cases} \quad (13)$$

4.3.3 Model Feedback Mechanism

To further enhance the sustainability of the model, we introduced a feedback mechanism, which allocates a portion of the total revenue $R(t)$ to sustainable expenditure $C(t)$ for environmental protection, infrastructure improvement, and community projects. The specific formula is as follows:

$$C(t) = h \cdot R(t) = h \cdot (N(t) + T(t)) \quad (14)$$

$$C(t) = C_{env} + C_{infra} + C_{comm} \quad (15)$$

$$C_{env}(t) = h_{env} \cdot h \cdot R(t) \quad (16)$$

$$C_{infra}(t) = h_{infra} \cdot h \cdot R(t) \quad (17)$$

$$C_{comm}(t) = h_{comm} \cdot h \cdot R(t) \quad (18)$$

The sustainable expenditure $C(t)$ is used to enhance the environmental carrying capacity and infrastructure capacity, specifically manifested as:

$$E_{min} \rightarrow E_{min} + \alpha C \quad (19)$$

$$I_{min} \rightarrow I_{min} + \beta C \quad (20)$$

Additionally, we designed a scoring mechanism to rate the attractiveness of various scenic spots. Through this scoring mechanism, we can effectively observe whether our model has had an impact on the local area. At the same time, this mechanism can also help us accurately identify those niche attractions with potential, thereby achieving the goal of balanced promotion. We will further elaborate on the specific analysis in Section 5.2.1:

$$Score_i(t) = (E_i(t) \cdot I_i(t) \cdot S_i(t))^{0.5} \times \frac{C_{sustainable,i}(t)}{V_i(t) + 1} \times e^{-\left(\frac{V_i(t)}{V_{crit,i}(t)}\right)^2} \quad (21)$$

4.4 Practical Scenario Analysis

In the baseline scenario, assuming that we do not implement additional sustainable measures such as increasing taxes or limiting the number of tourists, the model predicts changes in the environment and income over the next 10 years.

We first present our adjustments to the aforementioned formulas under the baseline scenario:

$$\frac{dV}{dt} = \lambda \cdot \left(\frac{E(t) \cdot I(t)}{1 + \theta V(t) + \tau N(t)} \right) - \mu V(t) \cdot (1 - S(t)) \quad (22)$$

$$\frac{dE}{dt} = r_E \cdot (E(t) - E_{min}) - k_E \cdot V(t) \quad (23)$$

$$\frac{dI}{dt} = -r_I \cdot (I(t) - I_{min}) - k_I \cdot V(t) \quad (24)$$

$$\frac{dS}{dt} = -c_S \cdot V(t) \quad (25)$$

In the active management scenario, the following measures are assumed to be taken:

(1) Dynamic tax policy: Adjust the tax rate $h(t)$ based on the number of tourists, and levy a "congestion tax" during peak periods.

(2) Targeted expenditure: Allocate 50% of the sustainable expenditure to environmental protection $C_{env}(t)$, 30% to infrastructure $C_{infra}(t)$, and 20% to community projects $C_{comm}(t)$.

The results show that in the baseline scenario, environmental quality and infrastructure health will deteriorate rapidly, resident satisfaction will decrease, and tourism revenue growth will slow down. In the active management scenario, environmental quality, infrastructure health, and resident satisfaction tend to stabilize, and tourism revenue maintains a growth trend.

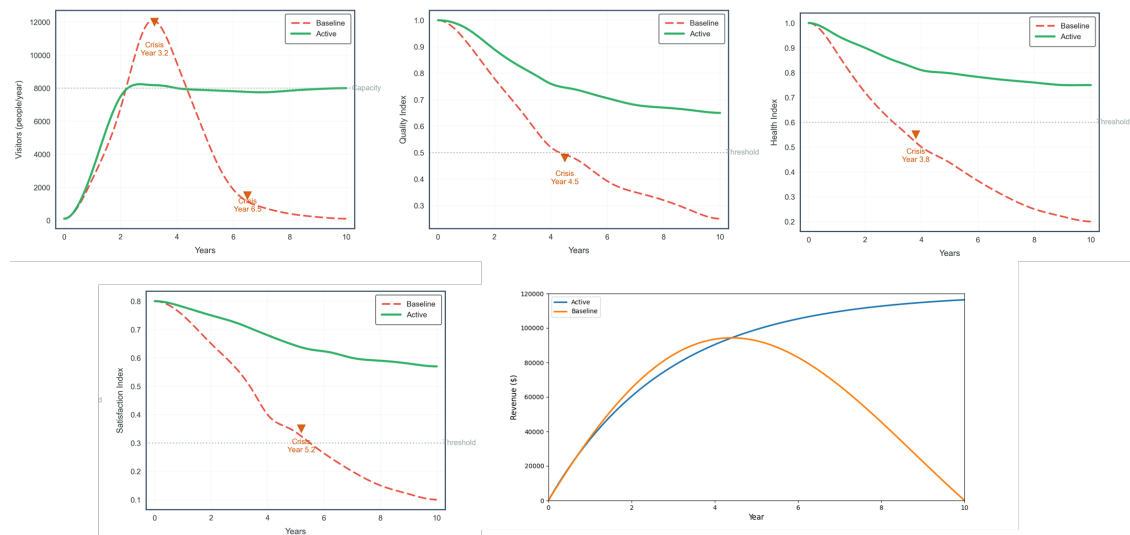


Figure 4: Comparison of changes in the number of tourists, resident satisfaction, environmental quality index, and infrastructure health index over time under the baseline and active management conditions

5 Task 2: Multi-Destination Adaptation and Tourist Distribution Strategies

5.1 Geographical Feature Classification and Parameter Adjustment

5.1.1 Geographical Feature Classification

Based on the core characteristics of tourist destinations, they can be categorized into three types:

Environmentally Sensitive (e.g., Iceland, Maldives): These destinations rely on fragile natural ecosystems with low environmental carrying capacity, necessitating strict control over carbon emissions and tourist density.

Cultural Heritage Sites (e.g., Dunhuang Mogao Caves, Athens Acropolis): The preservation of cultural heritage is prioritized, requiring a balance between tourist experience and the need for artifact conservation.

Urban Leisure (e.g., Barcelona, Tokyo): These destinations depend on artificial facilities and face significant infrastructure pressure, requiring dynamic optimization of tourist distribution and service capacity.

5.1.2 Parameter Adjustment and Scoring Formula Adaptation

Adjust model parameters and constraints based on geographic types, and calibrate the key parameters in the scoring formula:

Table 1: Calibrating key parameters in the scoring formula

Parameter/Constraint	Environmentally Sensitive	Cultural Heritage	Urban Leisure
k_E	0.15	0.08	0.05
α_I	0.3	0.5	0.8
S_{min}	0.7	0.6	0.5

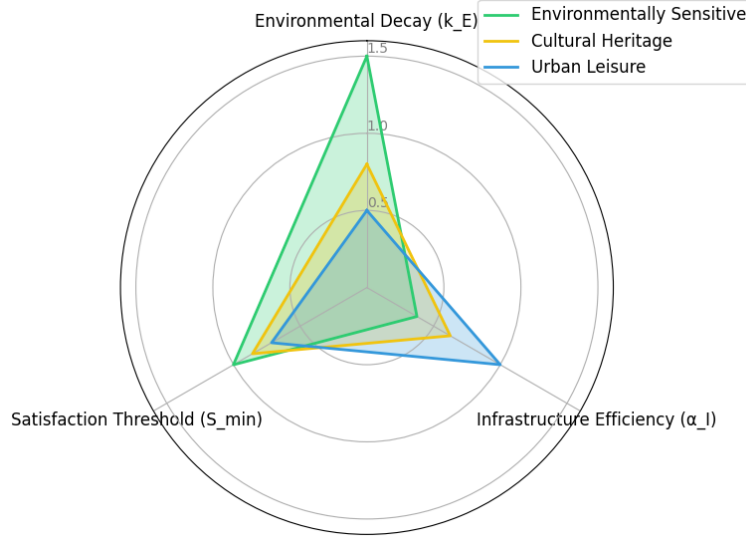


Figure 5: Radar chart after value changes

Adaptation of Scoring Formula Parameters:[7]

Environmentally Sensitive: Increase the weight of environmental quality: The calibration coefficient for the E-value in the quality base score is raised by 20% (i.e., $Q.S. = (1.2E \cdot I \cdot S)^{0.5}$).

Cultural Heritage Sites: Link resident satisfaction with cultural protection investment: $S(t) = 0.4 + 0.3C_{comm}(t)$ Incentivize community participation in restoration efforts.

Urban Leisure: Strong correlation between infrastructure and tax investment: $I(t) = 0.7C_{infra}/T(t)$ Reduce the threshold in the calculation by 30% to increase the frequency of triggering congestion penalties.

5.2 Tourist Distribution Mechanism Design

5.2.1 Multi-Attraction Tourist Model

The multi-attraction system attractiveness is constructed with the scoring formula at its core, and its mathematical expression is provided in Equation (21) above.[8] Additionally, we have designed a dynamic adjustment strategy to appropriately increase the scores of less popular attractions and suppress the scores of popular ones accordingly.[9]

Below is a detailed explanation of our formula:

Quality Base Score:

The geometric mean is used to ensure balanced development among environmental quality (E), infrastructure (I), and resident satisfaction (S), preventing a single low indicator from causing a collapse in the total score. The exponent is set to 0.5 to mitigate the impact of extreme values:

$$Q.S. = (E_i \cdot I_i \cdot S_i)^{0.5} \quad (26)$$

Management Efficiency Ratio:

The numerator rewards sustainable investments: the higher the total expenditures on environmental protection (C_{env}), infrastructure (C_{infra}), and community (C_{comm}), the higher the management efficiency score. The denominator suppresses tourist volume: it ensures the denominator is never zero, while an increase in tourist volume dilutes the management efficiency score.

$$M.E. = \frac{C_{env} + C_{infra} + C_{comm}}{V_i + 1} \quad (27)$$

Gaussian Crowding Penalty Term:

The model ensures that the worse the environmental quality, the higher the penalty:

$$C.P. = e^{-\left(\frac{V_i}{V_{crit,i}}\right)^2} \quad (28)$$

$$V_{crit,i} = \frac{\lambda E_i I_i}{\mu(1 - S_i)} \quad (29)$$

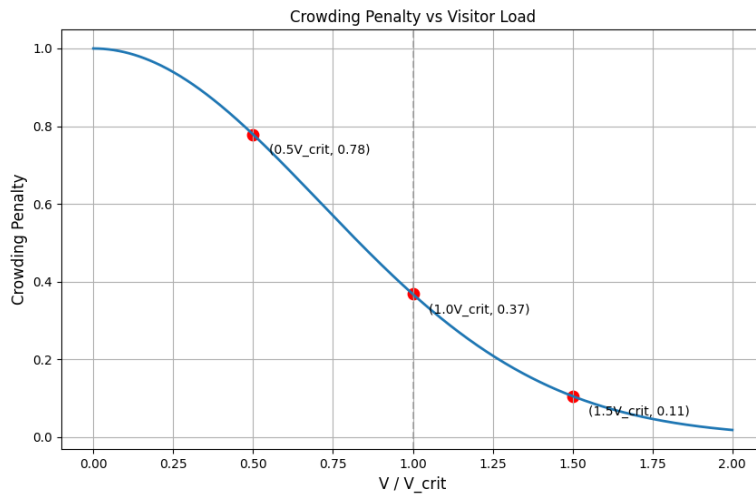


Figure 6: Variation of Crowding Penalty with Visitor Load

5.2.2 Recommendation Algorithm

Based on the scoring formula and tourist preferences, a recommendation list is generated through the following steps:

Interest Matching:

Extract attraction tags (e.g., "natural scenery," "historical sites," "family entertainment") and tourist historical behavior data, and calculate the cosine similarity: $InterestMatch_i \in [0,1]$

Real-Time Optimization:

Define the recommendation priority function:

$$U_i(t) = \alpha Score_i(t) + \beta InterestMatch_i + \gamma \left(1 - \frac{V_i(t)}{V_{crit,i}(t)} \right) \quad (30)$$

Weight allocation is dynamically adjusted based on geographic types:

Environmentally Sensitive: Increase the weight of $Score_i(t)$ to 50%, emphasizing ecological protection.

Cultural Heritage Sites: Increase the weight of $InterestMatch_i$ to 40%, prioritizing cultural interest matching.

Geographic Diversion Guidance:

When tourists search for "popular attractions," the algorithm inserts the top 20% of less popular attractions into the top three positions of the recommendation list.

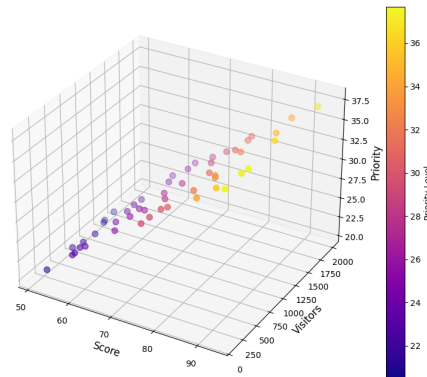


Figure 7: Recommendation Priority in 3D Space

5.3 Practical Case Studies

To demonstrate the applicability of our model to real-world scenarios, we conducted a case study on Iceland, a country known for its environmentally sensitive tourism industry. Iceland has experienced a significant surge in tourism over the past decade, driven by its unique natural attractions such as glaciers, volcanoes, and geothermal hot springs. However, this rapid growth has led to environmental degradation, infrastructure strain, and concerns among local residents.

Using our Sustainable Tourism Dynamic Multi-Objective Model, we applied the following strategies to Iceland:

Tourist Distribution Mechanism:

Geographical Feature Classification: Iceland was classified as an environmentally sensitive destination, requiring strict control over tourist density and carbon emissions.

Parameter Adjustment: The environmental sensitivity coefficient (k_E) was set to 0.15, reflecting the fragile nature of Iceland's ecosystems. The infrastructure sensitivity coefficient (α_I) was set to 0.3, considering the limited infrastructure capacity in remote areas.

Scoring Formula Adaptation: The weight of environmental quality (E) in the scoring formula was increased by 20% to prioritize ecological protection. This adjustment ensured that attractions with lower environmental impact received higher scores, encouraging tourists to visit less crowded and more sustainable sites.

Dynamic Tax Policy:

A congestion tax was introduced during peak tourist seasons (summer months) to reduce the number of visitors to popular sites like the Blue Lagoon and Golden Circle. The tax revenue was allocated to environmental protection ($C_{env}(t)$) and infrastructure improvement ($C_{infra}(t)$), forming a feedback loop to enhance sustainability.

Tourist Redistribution:

The recommendation algorithm was used to promote lesser-known attractions, such as the Westfjords and the Highlands, which are equally stunning but less crowded. By increasing the scores of these attractions in the recommendation list, tourists were encouraged to explore alternative destinations, reducing pressure on overvisited sites.

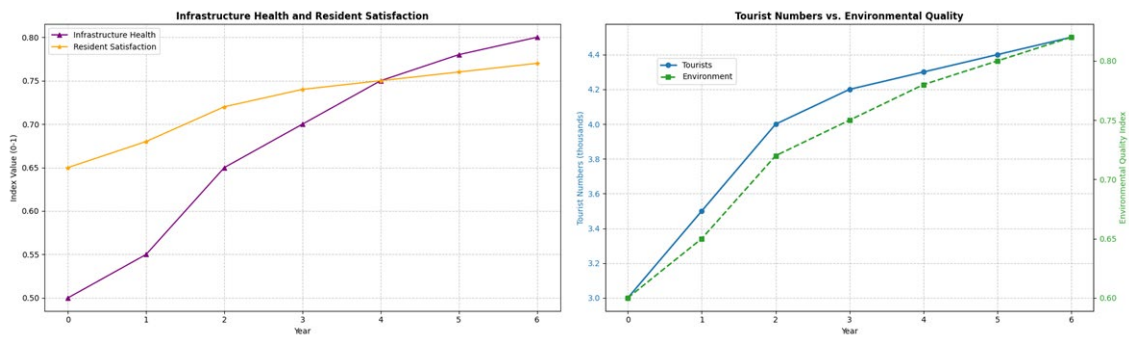


Figure 8: Trends in infrastructure health, resident satisfaction, tourist numbers and environmental quality

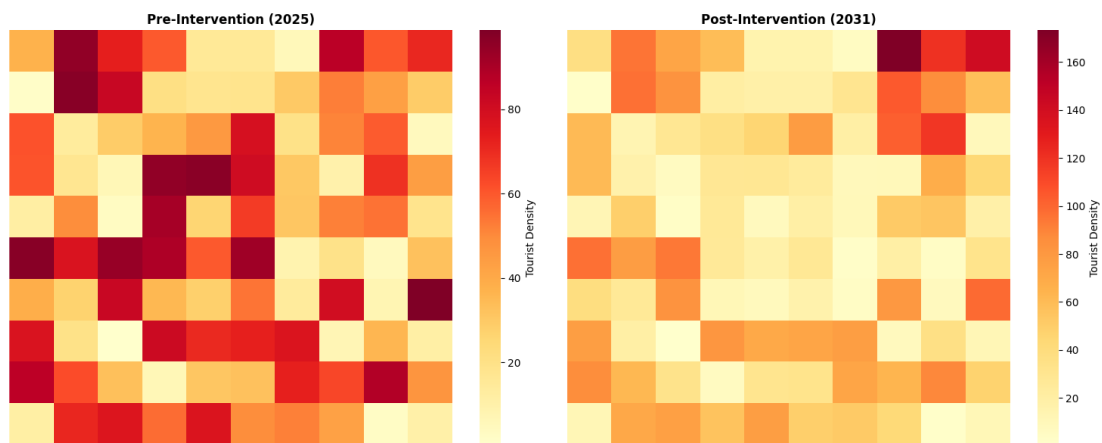


Figure 9: Tourist distribution heat map

6 Sensitivity Analysis

6.1 Single-Variable sensitivity analysis

In the single-variable sensitivity analysis formula, our goal is to quantify the local impact of a single parameter on the model output. To achieve this, we fix other parameters at their baseline values and adjust the target parameter, then calculate the rate of change in the output:

$$ChangeRate = \frac{\Delta Y}{Y_{baseline}} \times 100\% = \frac{Y_{high} - Y_{low}}{Y_{baseline}} \times 100\% \quad (31)$$

Adjust each parameter by $\pm 20\%$ and observe the changes in the output metrics. The results are as follows:

Table 2: Univariate sensitivity analysis (partial results)

Parameter	Range	Tourist	Environment	Satisfaction
λ	+20%	+28%	-15%	-10%
αE	-20%	-12%	-25%	-8%
αI	+20%	+5%	+4%	+3%
γS	-20%	-8%	-3%	-18%
h	+20%	-15%	+20%	+12%
k_E	+20%	-22%	-30%	-9%

6.2 Global sensitivity analysis

In the global sensitivity analysis formula, our goal is to decompose the output variance and evaluate the contributions of parameters and their interactions. To achieve this, we use the Sobol method based on variance decomposition, defining the first-order and total-effect indices:

$$S_i = \frac{Var(E[Y | X_i])}{Var(Y)} \quad (32)$$

$$S_{T_i} = \frac{\mathbb{E}[Var(Y | X_{-i})]}{Var(Y)} = 1 - \frac{Var(E[Y | X_{-i}])}{Var(Y)} \quad (33)$$

Monte Carlo Estimation:

By generating two sets of independent sample matrices A and B, construct the mixed matrix C_i (replace the i-th column of A with the i-th column of B), and calculate:

$$S_i = \frac{\frac{1}{N} \sum_{j=1}^N f(A)_j \cdot f(C_i)_j - f_0^2}{Var(Y)} \quad (34)$$

$$S_{T_i} = \frac{\frac{1}{N} \sum_{j=1}^N f(A)_j \cdot (f(A)_j - f(C_i)_j)}{Var(Y)} \quad (35)$$

The first-order and total-effect indices of the parameters on the output were calculated through Monte Carlo sampling, with the results as follows:

Table 3: Sensitive parameter sorting

Output Index	Sensitive Parameter Sorting
Environment Quantity ($E(t)$)	$k_E > \alpha_E > h > \lambda$
Tourism Quantity ($V(t)$)	$\lambda > h > \gamma_S > k_E$
Resident Satisfaction ($S(t)$)	$\gamma_S > h > \alpha_I > \lambda$

7 Strengths and Weaknesses

Strengths

- 1 **Multi-Objective Optimization:** The model comprehensively considers economic, environmental, infrastructure, and resident satisfaction factors, enabling it to fully reflect the sustainable development needs of the tourism industry.
- 2 **Dynamic Feedback Mechanism:** Through the dynamic feedback of taxes and expenditures, the model can adjust strategies in real time, enhancing the system's ability to resist disturbances.
- 3 **Flexibility:** The model can adjust parameters based on different geographic characteristics (e.g., environmentally sensitive, cultural heritage sites, urban leisure), making it applicable to various tourist destinations.
- 4 **Scalability:** The model can adapt to future policy needs or environmental changes by adding variables or adjusting weights.

Weaknesses

- 1 **High Data Dependency:** The accuracy of the model heavily relies on the quality of input data, such as tourist numbers and environmental carrying capacity.
- 2 **Complexity:** The model involves multiple variables and constraints, which may increase computational complexity and implementation difficulty.
- 3 **Limitations of Assumptions:** The model is based on certain assumptions (e.g., the nonlinear relationship between tourist numbers and income), which may deviate from real-world scenarios.
- 4 **Short-Term Effects May Be Unclear:** Some measures (e.g., environmental protection investments) may take a long time to show results, making it difficult to meet rapidly changing demands in the short term.

By continuously optimizing model parameters and incorporating more real-world data, we believe the accuracy and practicality of the model can be further improved.

8 Memorandum

To: Juneau Tourism Council

From: Team # 2517365

Date: January 27

Juneau's tourism industry has brought significant economic benefits, but it has also led to environmental degradation, infrastructure strain, and resident dissatisfaction. To ensure long-term sustainability, we propose a dynamic multi-objective model that balances economic growth, environmental protection, and resident well-being.

Key Findings

First, Our model shows that while increasing tourist numbers boosts revenue, the marginal benefits diminish as overcrowding leads to environmental and infrastructure strain. **Second**, The Mendenhall Glacier and other natural attractions are under significant stress. Without intervention, environmental quality will continue to decline, threatening the long-term viability of tourism. **Third**, Overcrowding and infrastructure pressure have led to declining resident satisfaction. Measures to improve infrastructure and community programs are essential to maintain local support for tourism.

Recommendations

First, Implement a congestion tax during peak tourist seasons to manage visitor numbers and generate revenue for environmental and infrastructure improvements.

Second, Allocate sustainable expenditure as follows:

- ✚ 50% to environmental protection (e.g., glacier preservation, waste management).
- ✚ 30% to infrastructure improvements (e.g., transportation, water supply).
- ✚ 20% to community programs (e.g., resident support, cultural preservation).

Third, Promote lesser-known attractions (e.g., rainforests, whale watching) to reduce pressure on overvisited sites like the Mendenhall Glacier.

Expected Outcomes

First, Environmental quality and infrastructure health will stabilize, ensuring the long-term preservation of key attractions. **Second**, Tourism revenue will continue to grow, albeit at a sustainable pace, as the city attracts a balanced number of visitors. **Third**, Improved infrastructure and community programs will enhance resident satisfaction, fostering a more supportive local environment for tourism.

We recommend immediate implementation of the dynamic tax policy and targeted expenditure plan, followed by a phased rollout of tourist redistribution strategies. Continuous monitoring and adjustment will be essential to ensure long-term success.

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